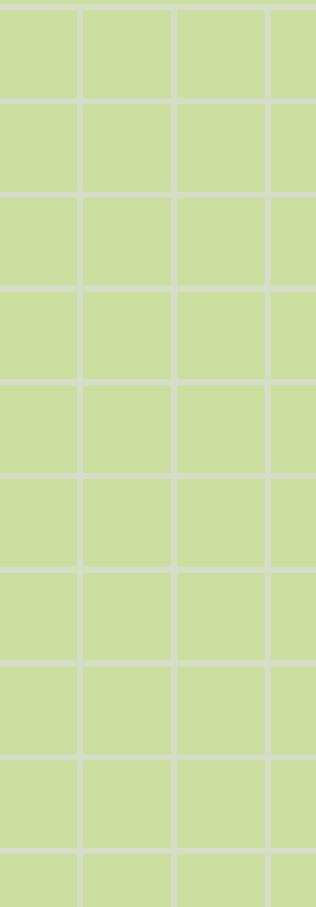




Basic Algebraic Properties of Complex Numbers



Addition

Multiplication

Closure

Adding or multiplying two complex numbers always gives another complex number. You never "leave" the set.

$$z_1 + z_2 = \text{complex} \quad | \quad z_1 \cdot z_2 = \text{complex}$$

Addition

Multiplication

Commutative

Order doesn't matter — flipping the two numbers gives the same result.

$$z_1 + z_2 = z_2 + z_1$$

$$z_1 \cdot z_2 = z_2 \cdot z_1$$

Addition

Multiplication

Associative

Grouping doesn't matter — you can bracket any two of the three numbers first.

$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

$$(z_1 \cdot z_2) \cdot z_3 = z_1 \cdot (z_2 \cdot z_3)$$

Additive identity

Zero does nothing

Adding $0 = 0 + 0i$ to any complex number leaves it unchanged.

$$z + 0 = z$$



Additive inverse

Negative cancels

Every complex number z has a negative $-z$.
Adding them gives zero.

$$z + (-z) = 0$$

Multiplicative identity

One does nothing

Multiplying any complex number by $1 = 1 + 0i$
leaves it unchanged.

$$z \cdot 1 = z$$

Multiplicative inverse

Reciprocal exists (except for zero)

For every non-zero complex number z , there's a number w such that their product is 1. We call w the **multiplicative inverse**, written z^{-1} .

$$z \cdot w = w \cdot z = 1 \implies w = z^{-1}$$

Multiplicative inverse

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Commutative Property Under Addition

1

Start — write z_1 and z_2 in standard form

$$z_1 = x_1 + iy_1 \quad z_2 = x_2 + iy_2$$

where x_1, x_2, y_1, y_2 are real numbers



2

Add them together

$$z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2)$$



3

Group real and imaginary parts

$$= (x_1 + x_2) + i(y_1 + y_2)$$



4

Swap order — real numbers can be added in any order

$$= (x_2 + x_1) + i(y_2 + y_1)$$

because $x_1, x_2, y_1, y_2 \in \mathbb{R}$ and real addition is commutative



5

Regroup back into complex form

$$= (x_2 + iy_2) + (x_1 + iy_1) = z_2 + z_1$$

Conclusion

$$z_1 + z_2 = z_2 + z_1 \quad \checkmark$$

Order doesn't matter when adding complex numbers.

Proof: Multiplicative Inverse

Finding z^{-1} for any non-zero complex number $z = x + iy$



1

Set up the inverse

Let $z^{-1} = u + iv$. By definition:

$$z \cdot z^{-1} = 1$$

2

Expand the product

Substitute and multiply out:

$$(x+iy)(u+iv) = 1+0i$$

$$(xu-yv)+i(xv+uy) = 1+0i$$

3

Match real & imaginary

Equate both sides:

$$xu - yv = 1$$

$$xv + uy = 0$$

4

Solve for u and v

Solve simultaneously:

$$u = x/(x^2+y^2)$$

$$v = -y/(x^2+y^2)$$

$$z \neq 0 \Rightarrow x^2+y^2 > 0$$

5

Write the inverse

Substitute u and v back:

$$z^{-1} = x/(x^2+y^2)$$

$$+ i \cdot (-y/(x^2+y^2))$$

Not defined when $z = 0$

Key Takeaways

- Every **non-zero** complex number has an inverse: $z^{-1} = 1/z$
- Division of two complex numbers ($z_2 \neq 0$) is written as z_1 / z_2
- The inverse helps us **divide** complex numbers easily.

Complex numbers obey these formulas



i

$$z^m \cdot z^n = z^{m+n}$$

ii

$$z^m / z^n = z^{m-n}, \quad z \neq 0$$

iii

$$(z^m)^n = z^{mn}$$

iv

$$(z_1 z_2)^m = z_1^m \cdot z_2^m$$

Thank You!